CROP DISEASE DETECTION

Early crop disease detection from a phone photo.

A leaf disease spotted a week early is a crop saved.
A grower holds up a phone; the model names the disease and shows the lesions it used to decide.

TEST ACCURACY

99.54%

on 1,087 held-out images

GPU

2.0 ms

per image, RTX 3050 Ti

CPU

6.8 ms

per image, batch 1

MODEL

3.2 MB

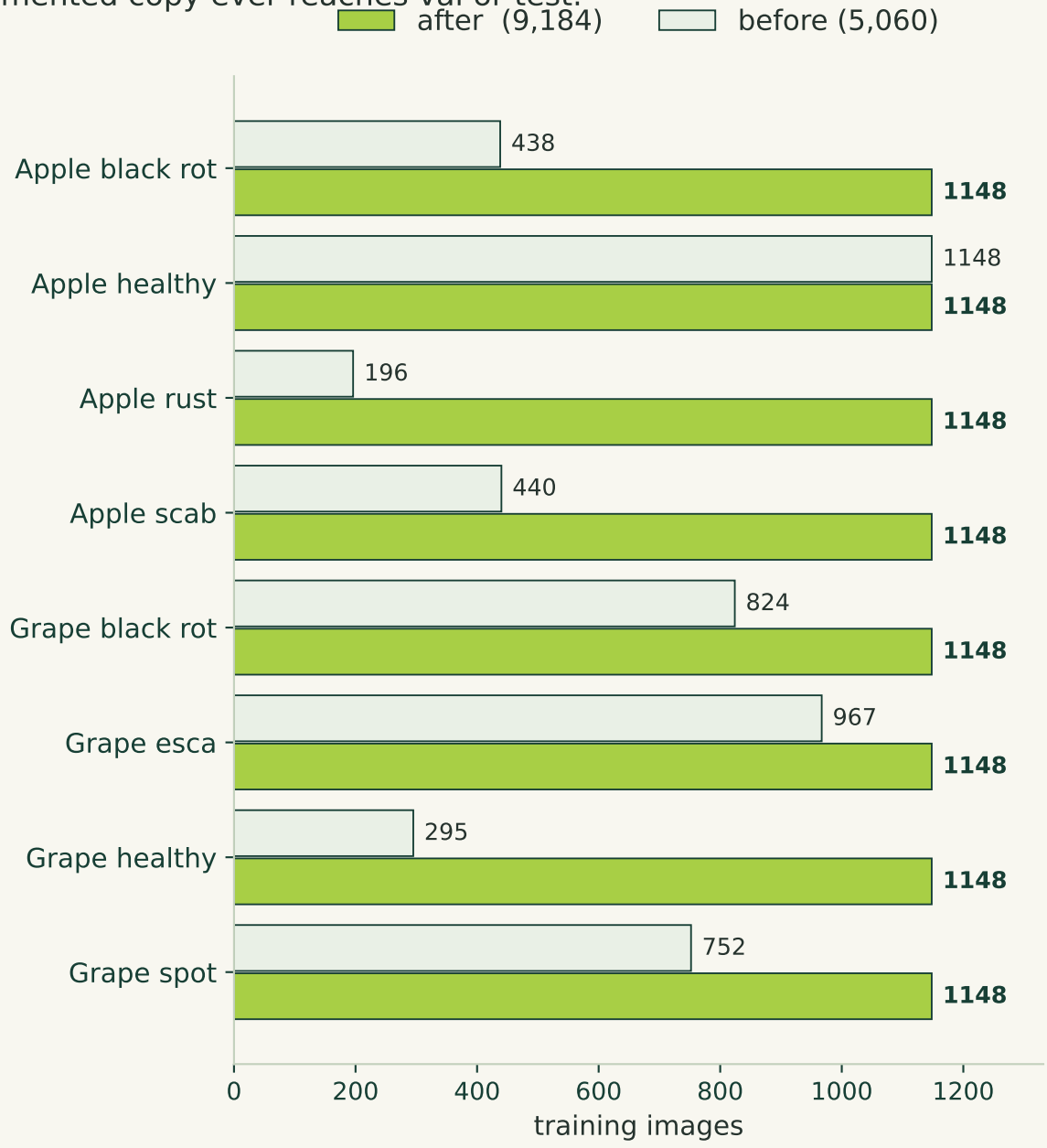
1.54M params, yolo11n-cls

8 classes - apple and grape - 7,233 source images - Grad-CAM explainability

STEP 1 - DATA

The dataset was lopsided

Train split before balancing: 196 to 1148 images per class, a 5.9x gap. Offline augmentation lifts every class to 1148. The split runs first, so no augmented copy ever reaches val or test.



Counted from dataset/train. Split seed 42, 70/15/15.

STEP 3 - PIPELINE

From photo to evidence

The leaf is segmented by an HSV threshold, then Grad-CAM shows which pixels drove the prediction. A heatmap on the background would mean a model reading the pot, not the plant.

photo



224x224 centre crop

leaf mask



HSV threshold, largest component

Grad-CAM



89% of the heat is on the leaf

Grape spot, predicted Grape spot at 100.0%. Colormap ivory to lime to terracotta.

STEP 4 - EXPLAINABILITY

Where the model looks

5 correct predictions and 1 real failure from the held-out test set. The heat marks the pixels that moved the logit.

Grape spot



100%

Apple black rot



100%

Apple healthy



100%

Apple scab



100%

Grape black rot



100%

Grape black rot



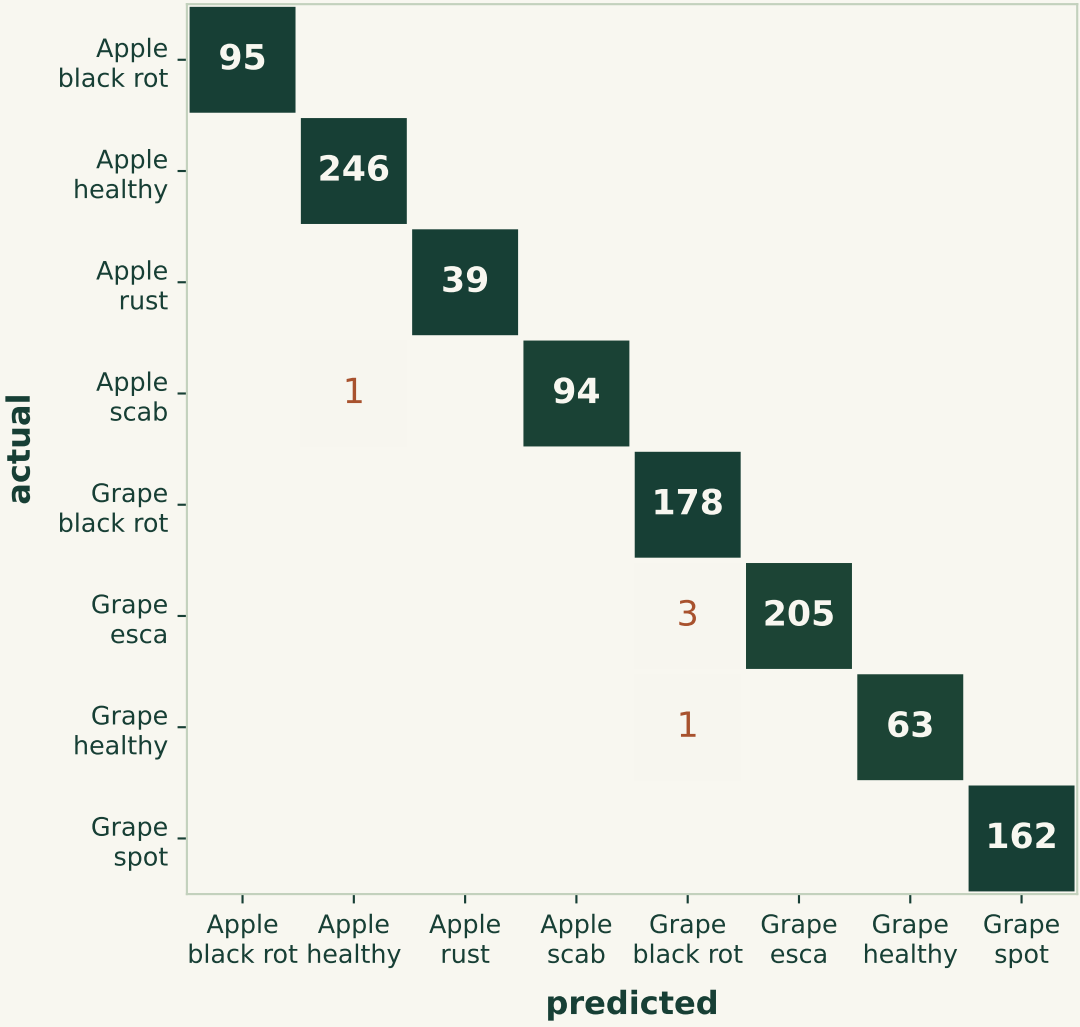
wrong, truly Grape esca

Heat inside the leaf mask: 64% to 89%, mean 76%.

STEP 5 - RESULTS

Confusion matrix, test set

1,087 held-out images, 5 wrong. Rows are the truth, columns are the prediction. Accuracy 99.54%.



Weakest class: Grape healthy, recall 98.4% on 64 test images.

yolo11n-cls, 224px input, seed 42. Test split only, never seen in training or validation.